

VIKRANT TOMAR

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RESEARCH PROFILE

I specialise in satellite remote sensing and in-situ observation of atmospheric composition, doing my PhD in Atmospheric Sciences from ARIES, expected in September 2026. My research work combines Level-2/Level-3 retrievals from TROPOMI, OMI, GEMS, and MODIS to detect and attribute emission sources, and quantifies decadal change in trace gases across major cities of South Asia, characterising diurnal and seasonal ozone formation regimes using hourly geostationary GEMS observations of NO₂ and HCHO. I have also utilised the oversampling technique on TROPOMI retrievals to resolve point sources NO_x hotspots and to map high-resolution ozone formation regimes over urban region to understand ozone sensitivity. I interpret space-based and ground-based observations together with atmospheric transport and chemistry modelling (HYSPLIT trajectory ensembles and receptor techniques, WRF, and box model), CAMS, ERA5 and MERRA-2 reanalysis, and EDGAR/CAMS emission inventories to understand atmospheric composition and air quality.

RESEARCH INTERESTS

Remote sensing of air quality; ozone-NO_x-VOC chemical regimes and FNR diagnostics; wildfire smoke and biomass-burning impacts on air quality; machine learning for satellite data applications; long-term trends in tropospheric ozone and its precursors.

EDUCATION

PhD in Atmospheric Sciences

Aug 2021 - present

Aryabhatta Research Institute of Observational Sciences (ARIES), registered with M.J.P. Rohilkhand University, Uttarakhand, India · Thesis submission expected September 2026

Master of Science: Physics and Astrophysics

Aug 2017 - Jun 2019

University of Delhi · New Delhi, India

Bachelor of Science: Physics

Aug 2014 - Jun 2017

Ramjas College, University of Delhi · New Delhi, India

RESEARCH EXPERIENCE

Satellite Remote Sensing of Atmospheric Composition

- **Geostationary diurnal analysis.** Analysing hourly GEMS retrievals of NO₂ and HCHO to quantify diurnal cycles and FNR-based ozone-regime transitions over the South Asian megacity integrated with surface ozone and PM_{2.5} observations.
- **Source detection through oversampling.** Oversampling technique of multi-year TROPOMI and OMI Level-2 retrievals into high-resolution maps that isolate persistent industrial and power-plant NO_x hotspots over South Asia.
- **Chemical formation regimes.** Analysis framework combining satellite retrievals of NO₂, HCHO and NH₃ columns (TROPOMI, GEMS, IASI) with ground observations to diagnose ozone and particulate formation regimes, including seasonal and spatial regime classification.
- **Decadal trend analysis.** Quantified two decades of change in tropospheric and columnar trace gases over major Indian urban regions from OMI, OMI/MLS, TROPOMI and MODIS records.

- **Fire and smoke analysis.** Used MODIS fire radiative power and fire-count products with satellite trace-gas columns to identify episodic biomass-burning sources and their influence on regional air quality downwind.

Atmospheric Modelling and Transport

- Ran and interpreted large **HYSPLIT back-trajectory** ensembles driven by GDAS meteorology, applying cluster analysis, concentration-weighted trajectory and potential source contribution function techniques to map source regions of observed trace-gas enhancements.
- **NCAR Master Mechanism** photochemical box model to evaluate ozone production and NO_x-VOC precursor sensitivity under observed conditions.

Ground-based Instrumentation and Field Campaigns

- **Remote-sensing instrumentation.** Pandora and MAX-DOAS spectroradiometers for columnar O₃, NO₂ and HCHO; FTIR spectrometer for CH₄ and CO₂ columns; DOAS retrieval principles and their link to satellite retrieval algorithms.
- **In-situ instrumentation.** Gas analyzers for O₃, NO_x, NO_y, CO and SO₂; low-cost sensor systems for PM_{2.5}, O₃, NO_x and CO; thermal-desorption GC-FID for hourly online NMHC measurements and offline campaign air samples.
- **Balloon-borne measurements.** Ozonesonde and radiosonde launches for vertical profiling of ozone and meteorology, providing profile references for satellite comparison.
- **Calibration and QA/QC.** Zero/span checks, multipoint calibrations with certified gas standards, flow calibration, leak checks and troubleshooting across the full analyzer suite.
- **Field campaigns.** Planned and executed trace-gas measurement campaigns across the Central Himalayas and the Indo-Gangetic Plain, including instrument deployment, site setup and logistics.

PHD RESEARCH — TROPOSPHERIC OZONE OVER SOUTH ASIA

- Investigated spatial and temporal variation in ozone and particulate matter and their chemical regimes over the Delhi megacity through a framework integrating ground-based ozone and PM_{2.5} observations with satellite-derived columnar NO₂, HCHO and NH₃.
- Quantified long-term (2007-2024) surface ozone variability and trends at a high-altitude Central Himalayan site by integrating ground-based measurements with satellite retrievals, regional model simulations and global reanalysis products.
- Determined the role of transported air masses in controlling surface ozone at the high-altitude site by combining transport modelling with receptor-oriented source-attribution techniques.
- Characterised variability of ozone, NO_x, NO_y, CO and NMHCs across the Central Himalayas and Indo-Gangetic Plain, assessing the influence of precursor emissions, biomass burning and regional transport.
- Assessed the impact of urbanisation on tropospheric ozone and its precursors across major Indian urban regions using two decades of satellite retrievals and reanalysis data.

PUBLICATIONS

Peer-Reviewed Publications

- Rajwar, M. C., Naja, M., Rawat, P., **Tomar, V.**, Kumar, M., Srivastava, P., & Lal, S. (2026). Non-methane hydrocarbon (NMHC) variability in rural Himalayan atmospheres: impacts on ozone, secondary organic aerosols, and human health. *Environmental Science and Pollution Research*, 1-14. <https://doi.org/10.1007/s11356-026-37694-z>
- Rajwar, M. C., Naja, M., Kant, Y., Rawat, P., **Tomar, V.**, Tiwari, R. K., & Lal, S. (2024). Ozone variability, its formation potential and crop losses in the Himalayan foothills. *Air Quality, Atmosphere & Health*, 1-14. <https://doi.org/10.1007/s11869-024-01572-9>

Submitted or Under Review

- **Tomar, V.**, Naja, M., Rawat, P., & Kumar, U. Spatiotemporal analysis of ozone and PM_{2.5} and chemical regimes over Delhi using multi-satellite and surface observations. (Under review, Urban Climate)
- **Tomar, V.**, Naja, M., Lal, S., Rawat, P., & Kumar, U. Long-term trends in air pollutants over major urban cities of India. (Under review, Atmospheric Environment)
- **Tomar, V.**, Naja, M., Kumar, M., Rawat, P., Lal, S., & Kumar, U. Long-term changes in ozone (2007-2024) at a high-altitude site in the Central Himalayas. (Under review, Elementa)

Conference Proceedings

- **Tomar, V.**, Naja, M., Lal, S., Rawat, P., & Kumar, U. (2024). Long-term changes in air pollutants at urban regions of India using remote sensing data. Proceedings of URSI-RCRS 2024. https://doi.org/10.46620/URSI_RSRC24/0994IDU7707
- Rajwar, M. C., Naja, M., Kant, Y., Rawat, P., **Tomar, V.**, Srivastava, P., & Lal, S. (2024). Unraveling ozone formation regimes in the Himalayan and IGP regions: A satellite-based study. Proceedings of URSI-RCRS 2024. https://doi.org/10.46620/URSI_RSRC24/0998APM8227

Manuscripts prepared and to be submitted soon

- **Tomar, V.**, Naja, M., Kumar, M., Rawat, P., Lal, S., & Kumar, U. Influence of regional transport on surface ozone in the Central Himalayas. In preparation.
- **Tomar, V.**, Naja, M., Kumar, M., Rawat, P., Rajwar, M., Lal, S., & Kumar, U. Surface ozone variability over the Indo-Gangetic Plain and Central Himalayas: Role of precursors, transport and biomass burning. In preparation.

TECHNICAL SKILLS AND DATA TOOLS

Programming	Python, R, MATLAB, C, C++, Fortran; Linux/HPC environments, Git/GitHub, LaTeX
Satellite products	TROPOMI/Sentinel-5P, OMI, OMI/MLS, GEMS (geostationary), AIRS, MODIS (AOD, fire radiative power, fire counts); Level-2/Level-3 NetCDF and HDF handling, QA and cloud filtering, regridding, collocation, oversampling
Models	HYSPLIT (trajectory clustering, CWT, PSCF), WRF, NCAR Master Mechanism photochemical box model
Reanalysis & inventories	ERA5, NCEP/GDAS, CAMS, MERRA-2; EDGAR and CAMS emission inventories
Ground-based remote sensing	FTIR (CH ₄ , CO ₂ columns), Pandora and MAX-DOAS (O ₃ , NO ₂ , HCHO columns), ozonesondes, Gas analysers (O ₃ , NO _x , CO)
Instrumentation & QA/QC	Gas analyzer calibration (zero/span, certified standards, flow checks), TD-GC-FID operation, instrument troubleshooting

PRESENTATIONS AND TRAINING

- **EGU General Assembly 2026** *May 2026*
Chemical Signatures and Sensitivity of Ozone and Particulate Nitrate from Multi-Satellite and Ground Observations over Northern India
- **AGU Annual Meeting 2025** *Dec 2025*
Chemical signature of Ozone and PM_{2.5} from space: A study over Northern India

- **7th Joint ECMWF/ESA/EUMETSAT Training on Atmospheric Composition** *Brussels, Oct 2025*
Royal Belgian Institute for Space Aeronomy: Satellite atmospheric composition products, data assimilation and modelling.
- **iCACGP-IGAC ECR Online Conference** *Sep 2025*
Ozone and particulate nitrate formation responses to precursors over a South Asian megacity: a multi-satellite approach integrated with ground-based observations.
- **Indian Clean Air Summit** *Bangalore, Aug 2025*
Ozone and PM2.5 chemical regimes over Delhi: a multi-platform observational study using satellite and ground-based networks.
- **IATA** *Dehradun, Oct 2024*
The persistent burden of air pollutants: a long-term trend analysis over India's major metropolises.
- **URSI-RCRS** *Bhimtal, Oct 2024*
Long-term changes in air pollutants at urban regions of India using remote sensing data.
- **iCACGP-IGAC Conference** *Kuala Lumpur, Sep 2024*
Decadal variabilities in tropospheric ozone over the Central Himalayas and major urban centers in India.
- **National Space Science Symposium** *Goa, Feb 2024*
Long-term changes in surface ozone over a high-altitude site in the Central Himalayas.
- **TROPMET** *Jaipur, Nov 2023*
Long-term changes in tropospheric ozone over the Central Himalaya: role of different processes.
- **iCACGP-IGAC ECR Online Conference** *Nov 2023*
Long-term changes (2007-2022) in surface ozone at a high-altitude site in the Central Himalayas.
- **5th ACAM Workshop** *Dhaka, Jun 2023*
Surface ozone at a high-altitude site in the Central Himalayas: long-term trends and role of different sources and factors.

AWARDS

- **CSIR-JRF 2020.** Council of Scientific and Industrial Research Junior Research Fellowship, All India Rank 124.
- **CSIR-JRF 2019.** Council of Scientific and Industrial Research Junior Research Fellowship, All India Rank 110.

MENTORING, ORGANIZING AND OUTREACH

- Mentored three master's and undergraduate students on a project using satellite remote-sensing and air-pollutant data to assess biomass-burning impacts on Delhi's air pollution (ARIES Training School in Observational Astronomy and Atmospheric Sciences, ATSOAA 2025).
- Served on the local organizing committee for URSI-RCRS 2024 and ATSOAA 2024 and 2025.
- Volunteered in science outreach and workshop organisation, including the 3rd BINA Workshop, AREA 2023, Aditya-L1 Science Support Cell Workshop 2023, Young Astronomers' Meet 2022, and ARIES National Science Day celebrations.

LANGUAGES

- **English.** Full professional proficiency, all research, publications and international presentations conducted in English.
- **Hindi.** Native speaker.